MILITARY SPECIFICATION

INTERROGATOR SET AN/TPX-46(V)7

This specification is approved for use by Electronics Research and Development Command, Department of the Army, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers the requirements for the construction and testing of an Identification Friend or Foe Interrogator Set designated AN/TPX-46(V)7.

2. APPLICABLE DOCUMENTS.

2.1 Issues of Documents. The effective issue or revision of the following documents shall be that listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplements thereto, unless (i) specific issues are set forth therefor in the cited specifications or (ii) issues different from those specified in the cited specifications are set forth in the solicitation. The date of the applicable DoDISS and supplements thereto shall be as specified in the solicitation or contract.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-P-11268	- Parts, Materials and Processes used in Electronic Equipment.
MIL-M-13231	- Marking of Electronic Items.
MIL-F-14072	- Finishes for Ground Electronic Equipment.
MIL-F-15024/6	- Plate, Identification, Information and Marking.

DOD AIMS SYSTEM PROGRAM OFFICE

DOD AIMS 64-900	- Mode 4 Input/Output Data KIR/KIT-1A/TSEC (FOUO). CEI Detail Specification, Part I.

STANDARDS

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-252	- Wired Equipment, Classification of Visual and Mechanical Defects.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command and Fort Monmouth, ATTN: AMSEL-EW-ES, Fort Monmouth, New Jersey 07703-5304, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5895

AMSC - N/A

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MIL-STD-454	-Standard General Requirements for Electronic Equipment.
MIL-STD-461	-Electromagnetic Emission and Susceptibility Requirement for the Control of Electromagnetic Interference.
MIL-STD-462	-Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-781	-Reliability Tests Exponential Distribution.
MIL-STD-810	-Environmental Test Methods.
MIL-STD-1472	-Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1562	-List of Standard Microcircuits.

DRAWINGS.

DL-SM-B-787682	-Interrogator Set AN/TPX-46A(V)7.
SM-A-787974	-Test Requirements for Signal Processor.
SM-A-787975	-Test Requirements for Control, Interrogator Set.
SPI 1G00132	-Special Packaging Instructions.

2.2 Other Publications.

Title 29 Code of Federal Regulations, Chapter XVII, Part 1910.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS.

3.1 First article. When specified (see 6.2) the contractor shall furnish first article units for first article inspection and approval (see 4.3).

3.2 Construction. The AN/TPX-46(V)7 shall be constructed in accordance with the requirements of this specification, DL-SM-B-787682, and the following list of major components:

Component	Drawing	Data list
Signal Processor, CP-1187A()/TPX-46(V)7	C5080975	DL-SM-B-787683
Control Interrogator Set C-9961()/TPX-46(V)7	SM-E-787795	DL-SM-B-787684
Cable assy, special purpose, control panel	SM-D-787797	DL-SM-B-787685
Cable assy, dc power	SM-D-787808	DL-SM-B-787692
Cable assy, ac power	SM-D-787809	DL-SM-B-787693
Installation Kit, program connector MK-1716()/TPX-46(V)7		

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3.2.1 Government furnished equipment (GFE). If required by contract, the following government furnished equipment shall be provided:

a. Computer KIR-1A/TSEC with Power Supply ZACA/TSEC and its characteristics specified in DOD AIMS 64-900 when supplied with primary power input of 115 Volts AC, 400 HZ.

b. Code Key KIF-18/TSEC.

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply microcircuits selected from MIL-STD-1562(see 4.4).

3.3.1 Recycled virgin and reclaimed materials. It is requested that, to the maximum extent possible, recovered or reclaimed materials, in lieu of virgin, shall be used without jeopardizing the intended use of the manufactured item.

3.4 Finish. All surfaces of each unit part of AN/TPX-46(V)7 requiring a protective coating shall be finished in accordance with MIL-F-14072 and the equipment drawings specified herein (see 4.4).

3.5 Nameplate and marking.

3.5.1 General. Marking of AN/TPX-46(V)7 shall be in accordance with MIL-M-13231.

3.5.2 Visibility. Whenever practicable, parts shall be so mounted that their identification marking will be readily visible with minimum disassembly of the equipment.

3.5.3 Nameplates. Each nomenclatured item shall have a nameplate conforming to MIL-P-15024/6, Type H or B (see 4.4).

3.5.4 Serial numbers. The following components of AN/TPX-46(V)7 shall be so serial-numbered (see 4.4):

Signal Processor, CP-1187(V)/TPX-46(V)7.

Control, Interrogator Set, C-9661()/TPX-46(V)7.

3.6 Safety engineering.

3.6.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection (see 4.13).

3.6.2 Corner and edge rounding. All exposed corners and edges shall be rounded to a minimum of 0.03 in (0.75mm) radius. Sharp edges and corners that present a personal safety hazard or potential damage to equipment during use shall be suitably protected or rounded to a minimum radius of 0.5 in (13mm).

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3.6.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses.)

3.6.4 System safety engineering. Each major component shall minimize the probability and severity of injury to personnel throughout the system's life cycle and be in accordance with Title 29, Code of Federal Regulations, Chapter XVIII, Part 1910, Occupational Safety and Health Standards, Requirement 1, of MIL-STD-454 and shall include, but not be limited to, paragraphs entitled "Labeling, Hazards and Safety" of MIL-STD-1472.

3.7 Interchangeability.

3.7.1 Mechanical interchangeability. Like units, assemblies, subassemblies and replaceable parts shall be physically interchangeable without modification and shall conform to Requirement 7 of MIL-STD-454 (see 4.8).

3.7.2 Electrical interchangeability. Like units, assemblies, subassemblies and replaceable parts shall be functionally interchangeable without modification and shall conform to Requirement 7 of MIL-STD-454. Electrical adjustments may be made at all maintenance levels, as referenced in the manuals (see 4.8.).

3.8 Dimensions and weights. The maximum equipment dimensions and weights shall be as listed in table I.

TABLE I. Dimensions and weights

Equipment	Height (in)	Depth (in)	Width (in)	Unit weight (lbs)
Signal Processor, CP-1187()/TPX-46(V)7	21.5 (54.6 cm)	22.12 (56.2 cm) (Excluding handles)	9.75 (24.3 cm)	85.0 (38.63 kg)
Control Interrogator Set, C-9661()/TPX-46(V)7	8.12 (20.6 cm)	16.0 (40.6 cm)	10.0 (25.4 cm)	10.0 (4.55 kg)

3.9 Environmental stress screening (ESS). The major component parts of the AN/TPX-46(V)7 shall meet the performance requirements specified herein after subjection to vibration and electrical preconditioning test of 4.6.

3.10 Performance requirements. The AN/TPX-46(V)7 shall meet the performance requirements as specified in 3.2 after subjection to the test of 4.9.

3.11 Service conditions. The AN/TPX-46(V)7 shall meet the following service conditions (see 4.10).

3.11.1 Altitude. The AN/TPX-46(V)7 shall meet specified performance, without degradation, at altitudes up to and including 10,000 feet (3048m) above sea level and shall withstand air transportation at altitudes up to 40,000 feet (12,192m) above sea level (see 4.10.1).

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3.11.2 High temperature. The AN/TPX-46(V)7 shall be operable without deterioration damage or degradation in specified performance at ambient air temperature as high as +125° F (51.67 °C) plus the effect of solar radiation at a rate of 360 BTU/ft 2/hr. The AN/TPX-46(V)7 shall be permitted a 30 minute cool-down period before full operation is required. At the end of the 30 minute period, the AN/TPX-46(V)7 shall provide full specified performance. The AN/TPX-46(V)7 shall be capable of withstanding, without degradation of specified performance, long periods of storage at temperatures as high as +160 °F (71.1° C) (see 4.10.2).

3.11.3 Low temperature. The AN/TPX-46(V)7 shall be operable without deterioration damage or degradation in specified performance at air temperatures as low as -50° F (-45.6° C). The AN/TPX-46(V)7 shall be permitted a 30 minute warm-up period; it may be turned on as desired to benefit overall operational performance. At the end of the 30 minute period, the AN/TPX-46(V)7 shall provide full specified performance. The AN/TPX-46(V)7 shall be capable of withstanding, without degradation of specified performance, long periods of storage at room temperature as low as -70° F (-56.7° C) (see 4.10.3).

3.11.4 Humidity. The AN/TPX-46(V)7 shall be operable without degradation in specified performance and shall sustain no physical damage, during and after prolonged exposure to extreme humidity levels as encountered in tropical areas (see 4.10.4).

3.11.5 Salt fog. After the salt fog test of 4.10.5, the AN/TPX-46(V)7 shall operate without degradation in specified performance and show no evidence of degradation, such as flaking, pitting, blistering, or loosening of finish or metal surface exfoliation. There shall also be no clogging or binding of moving mechanical parts.

3.11.6 Fungus. After the fungus test of 4.10.6, the AN/TPX-46(V)7 shall be visually examined. The AN/TPX-46(V)7 shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10 percent or less of the total area and no more than six unrelated minuted colonies with mycelial development in areas only in other than critical circuit portions, such as terminal spacing, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. In addition, the AN/TPX-46(V)7 shall operate without degradation in specified performance after exposure to the test conditions of 4.10.6. Materials used in the manufacture, fabrication, construction and treatment of the AN/TPX-46(V)7 shall comply with Requirement 4 of MIL-STD-454.

3.11.7 Rain. The AN/TPX-46(V)7 shall be operable without degradation in the specified performance after exposure to precipitation at variable rates to a total of 9.5 inches over a twelve hours period with a horizontal wind velocity of 35 knots per hour (see 4.10.7).

3.11.8 Dust. The AN/TPX-46(V)7 shall be operable without degradation in specified performance after being subjected to the dust test of 4.10.8.

3.11.9 Vibration, sinusoidal. The AN/TPX-46(V)7 shall operate within specified performance after being subjected to vibration test specified in 4.10.9.

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3.11.10 Shock. The AN/TPX-46(V)7 shall operate within specified performance after being subjected to the method and condition of the shock test specified in 4.10.10.

3.11.11 Bench-handling. The AN/TPX-46(V)7 shall withstand the shock as encountered during servicing. The equipment shall withstand this test removed from its normal enclosure as it would appear on the repair level (see 4.10.11).

3.12 Electromagnetic interference. When tested in accordance with 4.11, the equipment shall meet the emission and susceptibility requirements of MIL-STD-461, Notice 4, as defined hereinafter:

CE01	CS01	(b)RE02	RS02
CE02	CS02	RE02.1	(e)RS03
(a) CE03	CS03		
(b) CE04	(d)CS04		
(a)(b) CE05	CS06		
(c) CE06			

a. Testing shall be performed with the actual interface cable or its equivalent.

b. Conducted switching transients shall have amplitudes less than the values specified in figure 5 of MIL-STD-461, Notice 4. Radiated switching transients shall meet the requirements of paragraph 6.13.4 of MIL-STD-461, Notice 4.

c. In the key-up condition, the requirements of paragraph 6.4 of MIL-STD-461, Notice 4, apply. The local oscillator, however, shall meet the requirements level of signal (not exceed -67 dBm). In key-down condition the requirements of paragraph 6.4.3 of MIL-STD-461, Notice 4 apply. However, the 2nd, 3rd and 4th harmonics of the fundamental shall be relaxed by 10 dB.

d. The spurious response rejection level shall be 80 dB.

e. Requirement RS03 shall be performed over the frequency range of 10 kHz to 12.4 GHz using the following field intensity levels:

10 kHz to 1.9 MHz - 1 volt/meter
2 MHz to 1.99 GHz - 50 volt/meter
2 GHz to 12.4 GHz - 5 volt/meter

3.13 Reliability. The AN/TPX-46(V)7 shall have a specified (θ_o) mean-time-between-failure (MTBF) of 450 hours the lower test MTBF (θ_t) shall be 225 hours. The reliability requirements shall be demonstrated under the conditions specified in 4.12.

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3.14 Workmanship. The AN/TPX-46(V)7 shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs in MIL-P-11268 (see 4.13):

General requirements for plastic	Soldering and brazing
Wiring and cabling, including:	General requirements for securing of parts
Slack	Cleaning prior to welding
Protection	Process for welding
Clearance	Controls
Splicing	Self-tapping screws
Connections, general	Cleaning after welding
Grounding, general	Printed wiring assemblies
Shielding on wire and cable	

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3). (Does not include packaging.)

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspection of equipment before packaging (see 4.5).

d. Packaging inspection (see 4.14).

4.3 First article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish five first article units of the AN/TPX-46(V)7 for group A, B, C and D testing.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in table II and shall be performed in the order specified.

4.3.3 First article data. The first article test plan and test reports shall be as required in the contract.

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TABLE II. First article inspection

Inspections	Reqt para	Test para	Order of tests: Unit 1	Unit 2	Unit 3	Units 4-5
1. Inspections covered by subsidiary documents 1/	3.3 3.4 3.5	4.4	Inspection to be performed on all units.			
2. Environmental stress screening.	3.9	4.6	Tests to be performed on all units.			
3. Group A 1/	See table III.		Tests to be performed on all units.			
4. Group B 1/	See table IV.		Tests to be performed on all units.			
5. Group C.						
Altitude	3.11.1	4.10.1	1			
Low temperature	3.11.3	4.10.3	2			
High temperature	3.11.2	4.10.2	3			
Humidity	3.11.4	4.10.4	4			
Salt fog	3.11.5	4.10.5		3		
Fungus	3.11.6	4.10.6	6			
Rain	3.11.7	4.10.7		2		
Dust	3.11.8	4.10.8	5			
Vibration, sinusoidal	3.11.9	4.10.9			1	
Shock	3.11.10	4.10.10			2	
Bench-handling	3.11.11	4.10.11			3	
6. EMI	3.12	4.11		1		
Safety checklist 2/	3.6	4.13.1				
7. Group D Reliability	3.13	4.12				1

1/ The inspection 1 to 4, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements.

2/ Safety checklist tests shall be performed only once during first article testing.

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4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable documents as part of the inspection required by this specification and the inspection requirement specified in the contract.

Item	Where required
Parts, materials, and processes; general	3.3
Finish	3.4
Nameplate and marking	3.5

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.5.1 through 4.5.4, and 4.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspection in table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition, not defined herein in the contract, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels and AQLs indicated in table III.

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the government.

TABLE III. Group A inspection

Inspection	Requirement paragraph	Test paragraph	AQL Major	AQL Minor
Visual and mechanical	3.1	4.13	1%	4%
		4.14	1%	4%
System safety	3.6	4.13	1%	4%
Performance	3.10	4.9	1%	1/

1/ All electrical defects are major.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to Table IV and to the special inspection levels of table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The group B sample plans, for the AQL's listed in table III, shall be inspection level S-4.

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TABLE V. Group C inspection

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
Altitude	3.11.1	4.10.1
High temperature	3.11.2	4.10.2
Low temperature	3.11.3	4.10.3
Humidity	3.11.4	4.10.4
Dust	3.11.8	4.10.8
Fungus	3.11.6	4.10.6
Subgroup 2		
EMI	3.12	4.11
Salt fog 1/	3.11.5	4.10.5
Rain	3.11.7	4.10.7
Safety checklist	3.6	4.13.1
Subgroup 3		
Vibration, sinusoidal	3.11.9	4.10.9
Shock	3.11.10	4.10.10
Bench handling	3.11.11	4.10.11

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after their inspection before proceeding with subsequent inspection.

4.5.4 Group D inspection. This inspection shall consist of the test specified in table VI and shall be performed on units from lots which have been subjected to and met group A and group B inspections.

4.5.4.1 Sampling for group D inspection. Two samples of AN/TPX-46(V)7 shall be selected at random for group D inspection from every 50 units, or fraction thereof, produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection lot.

TABLE VI. Group D inspection.

Inspection	Requirement paragraph	Test paragraph
Reliability	3.13	4.12

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4.5.4.2 Group D failures. In case of group D failure occurrence, procedures and requirements specified elsewhere in the contract shall be followed.

4.5.4.3 Reinspection of group D sample units. Unless otherwise specified, sample units which have been subjected to and have passed group D inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection only.

4.6 Environmental stress screening. Each of the major components of AN/TPX-46(V)7 shall be subjected to vibration and electrical preconditioning. The preconditioning testing cycle shall consist of a four-hour period during which the chamber achieves a -45°C (-50°F) temperature (equipment not operating) followed by a four-hour period during which the temperature rises to +50°C (+124°F) (equipment operating). During the equipment operating period, the equipment shall be mechanically cycled periodically through its various phases of operation, as specified in the contract. Each equipment will be vibrated 10 minutes each operating hour at 1.25g ±10% peak acceleration at any nonresonant frequency between 20 and 30 Hz. When a failure occurs, the equipment shall be repaired and preconditioning shall continue until at least three preconditioning cycles have been completed, of which the last cycle shall be failure free. A failure is any condition which results in a malfunction in the equipment. Preconditioning time and failures shall be recorded for each equipment. After the preconditioning and prior to electrical acceptance inspection, the equipment shall not be repaired, aligned, cleaned or adjusted except to the extent possible by use of the external operating controls (see 3.9).

4.7 Weight and dimensions. The major components of AN/TPX-46(V)7 shall be weighed on a calibrated scale and measured to determine compliance with weight and dimension requirements of table I (see 3.8).

4.8 Interchangeability. The mechanical dimensions and electrical performance of parts shall be physically and electrically interchangeable among the units. Mechanical dimensions and electrical performance required by the drawings shall be measured to determine compliance with Requirement 7 of MIL-STD-454 (see 3.7).

4.9 Performance test. The AN/TPX-46(V)7 shall be subjected to performance and electrical inspection in accordance with SM-A-787974 and SM-A-787975 (see 3.10).

4.10 Service conditions. The service condition tests shall be performed as detailed below. Each major component of AN/TPX-46(V)7 subjected to these tests shall have passed group A tests and shall be resubmitted to and pass these group A tests after completion of all service condition tests. Where operation of the test item is required during any service condition test, the test item shall be required to pass an operational test of 4.9.

4.10.1 Altitude. The AN/TPX-46(V)7 shall be subjected to the test of MIL-STD-810, Procedure 1, Method 500.2. The AN/TPX-46(V)7 shall be tested at 40,000 Ft (12,192m) non-operating for one hour. The operating altitude in Step 7 shall be 10,000 Ft (3,048m). Inability of AN/TPX-46(V)7 to meet the requirements of 3.11.1 shall constitute failure of this test.

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4.10.2 High temperature. The AN/TPX-46(V)7 shall be subjected to the test of MIL-STD-810, Method 501.2, Procedure II. In Step 7 the temperature shall be 160°F (71°C). In Step 10 lower the temperature to 125°F (52°C), initiate operation of the equipment and at the end of a 30 minute period or when temperature stabilization occurs (whichever occurs first), the equipment shall show no degradation in performance (see 3.11.2).

4.10.3 Low temperature. The AN/TPX-46(V)7 shall be subjected to the test of MIL-STD-810, Method 502.2, Procedure 11. The temperature in Step 1 shall be maintained for a period of not less than 2 hours following stabilization of the test item. Following the 2 hour stabilization period, raise the chamber temperature to the low operating temperature (Step 4) of -50°F (-45.6°C) and maintain until temperature stabilization of the test item is reached. In Step 5, initiate operation of the equipment and at the end of 30 minute period or when temperature stabilization occurs (whichever occurs first), the equipment shall show no degradation in performance (see 3.11.3).

4.10.4 Humidity. The AN/TPX-46(V)7 shall be subject to the test MIL-STD-810., Method 507.2, Procedure II. Prior to measurements, excess surface water may be removed from the equipment by wiping external surfaces only. In Step 6, operation of 4.9 measurements shall be performed during the last 5 hours of the last cycle (see 3.11.4).

4.10.5 Salt fog. The AN/TPX-46(V)7 shall be tested in accordance with Method 509.2, Procedure 1, of MIL-STD-810. Inability of the AN/TPX-46(V)7 to meet the requirements of 3.11.5 shall constitute failure of this test.

4.10.6 Fungus. The AN/TPX-46(V)7 shall be subjected to the test of Method 508.3 of MIL-STD-810. There shall be abundant growth colonization on 50 percent or more of the control item after 14 and 28 days. No cleaning of the AN/TPX-46(V)7 is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the AN/TPX-46(V)7. Inability of the AN/TPX-46(V)7 to meet the requirements of 3.11.6 shall constitute failure of this test.

4.10.7 Rain. The AN/TPX-46(V)7 shall be tested in accordance with Method 506.2, Procedure 1, of MIL-STD-810. The AN/TPX-46(V)7 shall be operated during the last 10 minutes of each 30 minute test period. In each 30 minute rain period, the wind shall be maintained for 15 minutes, commencing 5 minutes after the start of the period (see 3.11.7).

4.10.8 Dust (fine sand). The equipment shall be subjected to the test of Method 510.2, Procedure 1, of MIL-STD-810, except that the air velocity shall be 200 ±100 ft/min (1 ±0.5 m/sec) in Step 2. Step 3 shall proceed immediately after temperature and humidity are stabilized. The test item shall be reoriented every 1.5 hours during Step 4. If the time required to reorient the test item exceeds 5 minutes, it shall not be included in the test time of 6 hours (see 3.11.8).

4.10.9 Vibration, sinusoidal. The equipment shall be subjected to the test of Method 514.3, Category VIII, Figure 514.3-6, Paragraphs 4.5.1.1 and 4.5.1.3, Procedure I, of MIL-STD-810. Test level shall be ±1.5g from 5.5 to 500 Hz (non-operating). Total vibration time shall be 3 hours, including resonance search time. Resonant dwell is not required (see 3.11.9).

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4.10.10 Shock. The equipment shall be subjected to the test of Method 516.3, Procedure 1, of MIL-STD-810. The shock pulse shall be at a nonoperating peak value of 20g's for a duration of 18 ms and an operating peak value of 9g's for a duration of 30 ms. (See 3.11.10.)

4.10.11 Bench-handling. The equipment shall be subjected to the test of Method 516.3, Procedure VI, of MIL-STD-810. (See 3.11.11.)

4.11 Electromagnetic interference. The AN/TPX-46(V)7 system and the individual components shall be tested for compliance to the requirements of MIL-STD-461 enumerated in 3.12. Tests shall be in accordance with the approved measurement techniques of MIL-STD-462, Notice 3. (See 3.12.)

4.12 Reliability. To determine compliance with 3.13, the equipment shall be tested in accordance with Plan IIIC of MIL-STD-781. The test level shall be in accordance with 4.12.1.

4.12.1 Reliability test conditions. The reliability tests described above shall be conducted under conditions prescribed by Test Plan IIIC, MIL-STD-781, except that vibration shall be at 1.25g's. Voltage cycling shall be excluded and on-off cycling shall be 23 hours on and 1 hour off. The following requirements of MIL-STD-781 shall also apply:

a. Detail reliability test procedures (5.1.3).

b. Inspection (5.2).

c. Test facilities (5.4).

d. Selection of equipment (5.5.1).

e. Installation of equipment (5.5.2).

f. Failure categories (5.6.1).

g. Verifying repair (5.8).

4.13 Visual and mechanical. The AN/TPX-46(V)7 shall be examined for the defects listed in MIL-STD-252. In addition, AN/TPX-46(V)7 shall be visually inspected to determine conformance to the requirements of 3.6.

4.13.1 Safety checklist. Using SEL 1183 checklist verify that the requirements of 3.6 are met.

4.14 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

b. Quality conformance inspection of packaging.

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4.14.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.14.2 Quality conformance inspection of packaging.

4.14.2.1 Material inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.14.2.2 Preservation inspection. Inspection of preservation and interior marking shall be in accordance with group A and group B quality conformance inspection requirements of MIL-P-116: Lot formation and sampling procedures shall be as specified therein.

4.14.2.3 Packing inspection. Inspection of packing and marking for shipment and storage shall consist of the examination specified in table VII. Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

TABLE VII. Packing inspection provision

No.	Characteristic	Method of inspection
101	Intermediate container not as specified.	Visual
102	Improper closure of intermediate container.	Visual
103	Shipping container not in accordance with specification.	Visual
104	Excessive cube	Visual
105	Improper block and bracing.	Visual
106	Closure not in accordance with specification.	Visual
107	Weight and size exceed container limitations.	Weight and measure
108	Strapping not in accordance with specification, incorrectly applied, or omitted.	Visual
109	Marking omitted, incorrect, or illegible.	Visual

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5. PACKAGING.

5.1 Packaging requirements. The requirements for packaging shall be in accordance with SP1 1G00132.

6. NOTES

6.1 Intended use. Interrogator Set AN/TPX-46(V)7 is intended for use with the Air Defense Radar Systems to provide the operator with a rapid and cryptographically secure means of identifying friendly aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or B preservation and packing (see Section 5).

c. When first article inspection rough handling tests are not required.

d. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

e. Marking and shipping of samples.

f. Place of final inspection.

g. Number of first articles to be submitted for approval (see 3.1).

h. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example, AN/TPX-46(V)7. The contractor should apply for nomenclature in accordance with the applicable provision in the contract (see 1.1).

6.5 Level B preservation. When Level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

Custodians:
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Preparing Activity:
Army -ER
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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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